AI BUILD LOG · RAZORPAY × ISB AI PM BUILD CHALLENGE · TRACK 2

How AI built Margin-Aware Cart Recovery, and where I decided

One builder, about three weeks, with Claude as researcher, critic and engineer. AI did the volume work; the product calls below are mine. Anuraag Praharaj, PGP Co'27.

Tools used

Claude (Anthropic)
Chat with a project knowledge base: the brief, JDs and course notes. Web search, code execution and artifacts.

Claude agentic workspace
Wrote and tested the prototype, deck and note; ran a headless browser to check every screen and slide.

Open-source libraries
Chart.js (charts), SheetJS (Excel), pdf.js (PDF upload parsing), Playwright and jsdom (testing).

GitHub + Vercel
Hosting the prototype and deck. No external APIs; all merchant data synthetic.

What AI did at each stage, and how it sped things up

STAGE	WHAT AI DID	ACCELERATION
Research	Read Razorpay's Magic Checkout docs (coupons, abandoned-cart webhook payload), Agent Studio launch and guardrail posts, and press coverage; separated what's public from what's assumed, labelling every claim factual / somewhat factual / guess	Hours instead of days; caught that the cart agent already picks among preset offers, which sharpened the gap to **profit + proof**
Ideation and critique	Stress-tested ideas I later dropped (Tatkal-style auctions, payment-data ad targeting, cross-merchant price advice) and V1 → V2 of this one	Every idea got a devil's-advocate pass before I committed time
Modelling	Built a synthetic world: 3 merchants, about 14k carts, 10 signals, response curves; calibrated it and computed flat / tiered / agent results	Numbers come from running the model, not guesswork
Prototype	Wrote a 6-screen Agent Studio-style app: Setup with Excel/PDF upload, Recommendations with next best actions, Response curves, Cart log, Performance, plus live-AI Q&A	Rebuilt three times in a day as my thinking changed; every screen auto-tested for errors
Deck and note	Pulled Razorpay's Blade design tokens (fonts, palette) from its public package; laid out 5 slides and this note; checked overflow on every slide	Design system matched without guessing

Key decisions I made

- **Track 2**, scoped to **abandoned carts**, with **profit, not conversion**, as the objective.
- **V2 scope:** only the merchant's **top-5 margin categories**, with the discount applied to those items only in mixed carts.
- **Experiment design:** random discounts in **0.5% steps** from 0 to the cap, with a fixed **zero-discount** group.
- Show offers in **rupees**, so odd percentages don't look arbitrary to shoppers.
- **Signal set:** split payment into UPI / cards / net banking / COD; time into slots; added region tiers, cart size relative to each merchant's AOV, failure reason, coupon behaviour and UTM source.
- **Bucket-level only:** no shopper histories; **excluded income and device**.
- Framed the **checkout tree** (α straight-through vs c(d) recovered) and the profit-vs-conversion objective.
- **New merchants** start from Razorpay's existing pooled data; the merchant **keeps the cap** for explainability.
- Put the merchant's **next action upfront** in the prototype; added a plain-language "How it works".

Where I overruled or corrected the AI

- AI suggested a few coarse test levels (0 / 5 / 10%). **I chose fine 0.5% steps**, and we fitted a curve so it stays statistically workable.
- AI proposed "drop-off stage" and "past coupon use" as signals. **I cut both**: the first isn't reliably available, and the second needs customer-level data.
- I **fact-checked the deck's claims**: "one coupon for every cart" became "a preset menu, flat or tiered", and "who gets paid" was reworded precisely.
- AI corrected me too: **hashing isn't encryption** and doesn't solve data sharing (contracts and opaque IDs do), and 31 category combinations can't each be tested, so we use one joint model.

What I'd tell a reviewer

AI let one person cover research, modelling, a working prototype and a branded deck in the time a team would take to scope it. What it couldn't do was decide what mattered: the objective, the scope, what to leave out, and where the privacy line sits. Those calls are the product judgment I'm submitting.

Prototype: https://margin-aware-cart-recovery.vercel.app · Deck: https://margin-aware-cart-recovery.vercel.app/deck · Product note submitted separately. Concept only, not a Razorpay product.